# Pet Hamster: Coding a micro:bit Pet

Year 5/6 STEAM | CRT lesson plan | 45 minutes | MakeCode in the micro:bit iPad app

**At a glance:** Students work in **pairs** with one **iPad**, one **micro:bit (V2)** and one **battery pack**. They follow the built-in MakeCode tutorial **Pet Hamster** to code Cyrus, a sleepy hamster who smiles when the gold logo is touched and gets sad when shaken. This is only their **second MakeCode lesson**, so keep instructions short and let the tutorial do the teaching. Downloading to the real micro:bit is a bonus, **not essential**. Fast finishers move to the Extension Challenge.

## Learning Intention and Success Criteria

**Learning Intention:**

To use inputs and events in code to make a micro:bit pet react to the way we interact with it.

**Success Criteria:**

I can:

- use **event blocks** (on start, on logo pressed, on shake) to start my code when something happens.
- use **output blocks** (show icon, play melody) to give Cyrus faces and sounds.
- **test** my code in the simulator and fix it if Cyrus does not react the way I planned.

## Key Words

| Word | Student-friendly meaning |
|---|---|
| **Input** | Something the micro:bit senses, like a touch, a shake or a sound. |
| **Output** | Something the micro:bit does, like showing a face or playing a sound. |
| **Event** | A block that waits for an input, then runs the code inside it. |
| **Simulator** | The on-screen micro:bit in MakeCode. It runs the code without the real device. |
| **Debug** | Finding and fixing a mistake in the code. |

## Lesson Sequence

| Time | Phase | What to do |
|---|---|---|
| **0 to 5 min** | Hook | Students sit in their pairs. Hold up a micro:bit and ask: *"If this micro:bit was a pet, how could it tell how you are treating it?"* Take 3 or 4 ideas. Point out the sensors students will use: the **gold logo** (touch) and the **accelerometer** (shake). Introduce Cyrus the hamster. |
| **5 to 12 min** | Explicit teaching | Read the LI and SC aloud. Explain the key words: an **input** is something the micro:bit senses, an **output** is something it does, and an **event** block runs its code only when that input happens. Show students how to open the tutorial using the **How to Open the Tutorial** screenshots. Then hand out one iPad, micro:bit and battery pack per pair. |
| **12 to 35 min** | Pairs code and test | Pairs complete the tutorial (about 6 short steps) and **test in the on-screen simulator**. Pairs can download to the real micro:bit if they want, but this is a bonus, not a requirement. Roam with the questions below. Finished pairs move to the Extension Challenge. |
| **35 to 41 min** | Share and reflect | Two or three pairs demo their Cyrus (on screen or on the micro:bit). Ask the class: *"Which block is the input? Which is the output? What made Cyrus go back to sleep?"* Quick journal entry: students draw Cyrus's **three faces** and label the input that causes each. |
| **41 to 45 min** | Pack up | **Unplug battery packs** from every micro:bit. Return iPads, micro:bits and batteries to their tubs. Check the count before students leave. |

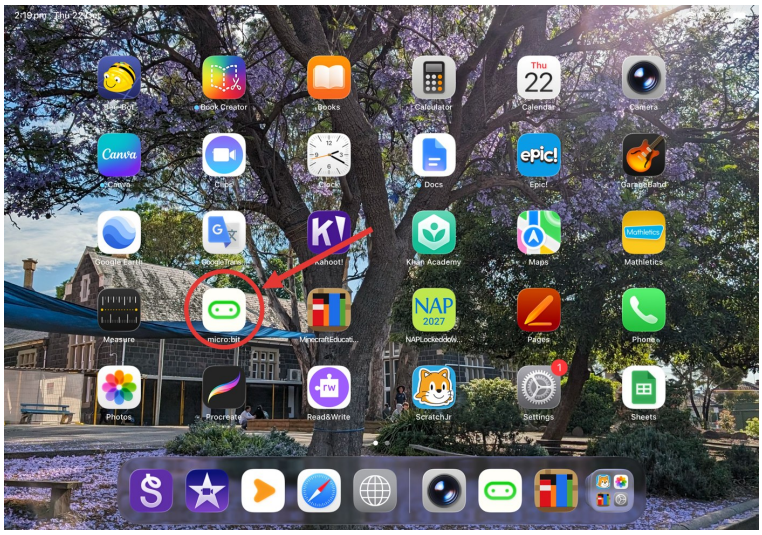
## Questions to Ask While Roaming

- Point to the block that is listening for an **input**. What is it waiting for?
- What would happen if you deleted the last **show icon [asleep]** block? Predict, then test it.
- What other ways could a real pet react? Which micro:bit sensor could detect that?

## How to Open the Tutorial

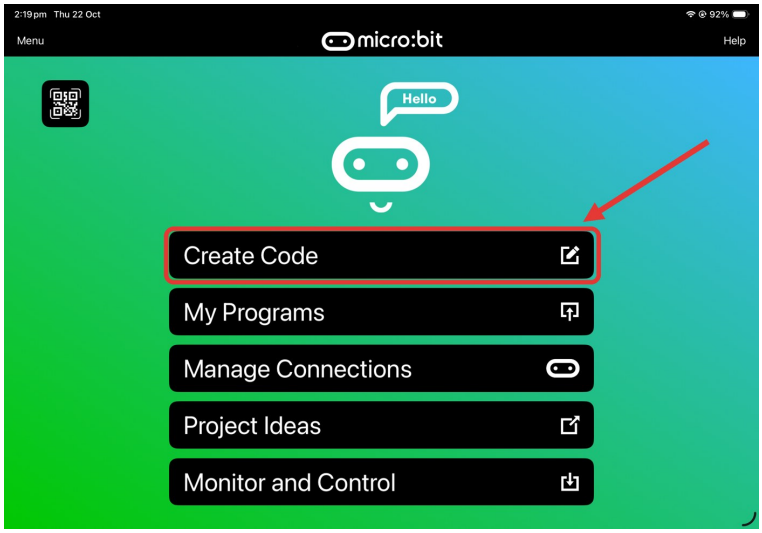
### 1

Open the **micro:bit** app on the iPad home screen.



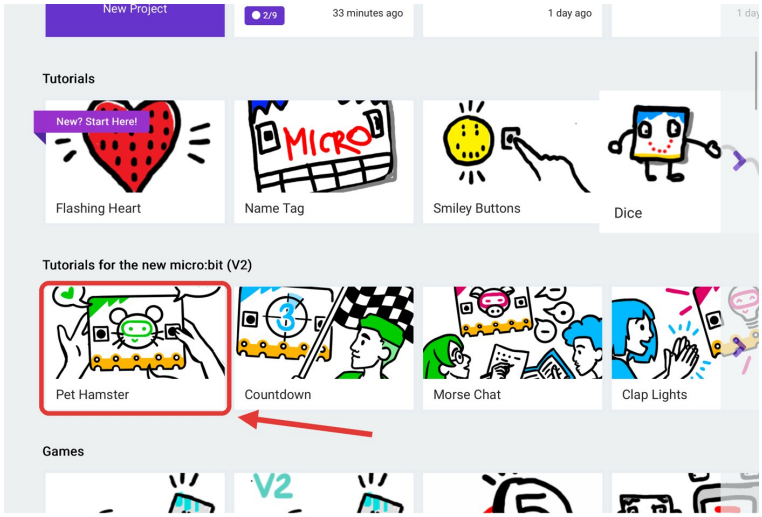
### 2

Tap **Create Code**. This opens MakeCode.




### 3

Scroll to **Tutorials for the new micro:bit (V2)** and tap **Pet Hamster**. The tutorial guides students one step at a time.



## What the Finished Code Looks Like

Use this to check student work at a glance. Block names match the MakeCode menus.

```
on start
    show icon [asleep -_-]
on logo [pressed]                       (Input menu)
    show icon [happy :)]
    play melody [jump up] [until done]   (Music menu)
    show icon [asleep -_-]
on [shake]                              (Input menu)
    show icon [sad :(]
    play melody [wawawawaa] [until done]
    show icon [asleep -_-]
```

**Using the real micro:bit:** downloading is **not essential**. The simulator shows everything students need, and they can tap the logo and press the SHAKE button on screen. If a physical micro:bit is not working, **unplug and replug the battery pack** and/or **swap to another battery pack**. If it still will not work, carry on in the simulator.

## Support

| Strategy | How it looks |
|---|---|
| **Model the first step** | Sit with the pair and do step 1 together (show icon asleep in on start). Then let them try step 2 on their own while you watch. |
| **Shrink the goal** | Success is **one working event** (logo makes Cyrus happy) tested in the simulator. Shake can come next if time allows. |
| **Colour matching** | Remind students that block colours match the menu: **Basic** is blue, **Input** is pink, **Music** is orange. Find the colour, then find the block. |
| **Buddy pair** | Pair a struggling pair with a finished pair. The finished pair explains, but the struggling pair keeps their own hands on the iPad. |

## Extension Challenge: Give Cyrus More Reactions

Once the tutorial is finished and tested, students add new reactions to Cyrus:

- Add **on button A pressed** (Input) so Cyrus shows a **heart**, then goes back to sleep.
- Add **on button B pressed** so Cyrus says their name using **show string** (Basic).
- Drag in another **on [shake]** block and change the dropdown to **tilt left**, **tilt right** or **screen down**. What would Cyrus do if they were turned upside down?